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ORIGINAL RESEARCH

Temporal brain tumor progression tracking using deep learning and 3D MRI volume analysis

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Abstract Cancer is among the most prevalent diseases globally. Concurrently, advances in artificial intelligence are revolutionizing brain tumor diagnosis by offering greater consistency and improved accuracy. This research introduces NeuroSight, a novel technique for tracking brain tumor progression. NeuroSight employs advanced deep learning algorithms to detect, segment, and quantify brain tumor volume from 3D MRI scans. The method combines multi-class EfficientNet and AGSE-VNet models for precise tumor detection and segmentation. NeuroSight's progression tracker subsequently measures and contrasts the segmented tumor regions over time for individual patients. The efficacy of NeuroSight was tested through a series of experiments. The obtained results show 31.3% improvements in the F1-score using EfficientNet. Also, the AGSE-VNet component achieves 2% and 155.8% better Dice Coefficient and Hausdorff Distance than alternative methods.

Keywords Brain tumor · Cancer detection · Tumor progression tracking · Deep learning · 3D Volume analysis

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1 Introduction

In 2023, the United States reported 1.9 million new cancer cases and 609,820 cancer-related deaths, with brain tumors constituting 2.17% of all cancer deaths and having a notably low five-year survival rate [1]. Untreated early-stage tumors can be fatal, emphasizing the challenges in neuro-oncology. Accurate monitoring of tumor progression is crucial, and while magnetic resonance imaging (MRI) is essential, manual analysis is time-consuming and prone to discrepancies. The pressing need for automated, consistent, and standardized MRI image analysis in neuro-oncology is underscored. Advances in artificial intelligence hold the promise of transforming brain tumor diagnosis, offering a more consistent and accurate methodology [2, 3].

This paper presents a new method for brain tumor progression called NeuroSight. NeuroSight leverages a deep learning system, integrating multi-class EfficientNet and AGSE-VNet models. EfficientNet model is used for image classification, while the AGSE-VNet model is employed for tumor segmentation. By incorporating these models, NeuroSight offers a novel method for detecting, segmenting, and volumetric measuring brain tumors using 3D MRI scans. To the best of our knowledge, NeuroSight is the first brain tumor progression tracking method based on tumor volume analysis over multiple time intervals. The contributions of this research paper are: (1) Leverage the multi-class EfficientNet to develop a brain tumor classification model. (2) Leverage the AGSE-VNet to develop a brain tumor segmentation model. (3) Propose a brain tumor progression tracking method based on tumor volume analysis over multiple time intervals.

The rest of this paper is organized as follows. Section 2 reviews the background and the related works. Section 3 introduces the proposed method, NeuroSight. Section 4

presents the evaluation setup of NeuroSight and the comparison results with baseline methods. Finally, the conclusion and some future directions are presented in Sect. 5.

2 Background and related works

Brain tumor diagnosis relies on MRI for soft tissue contrast and multi-planar imaging. Despite MRI advantages, tasks like segmentation, classification, and progression monitoring are complex and prone to variability, highlighting the need for advanced analytical tools. Radiomics, extracting features from images through high-throughput methods, utilizes machine learning to reveal hidden tumor heterogeneities. Integration into clinical practice faces challenges, including reliable image preprocessing and feature extraction methods. Traditional imaging methods face limitations, including subjectivity in interpretation, radiologist fatigue, and difficulties in capturing the diversity of brain tumors at the cellular and molecular levels.

ML shows promise in addressing these challenges. Several contributions have been made in the literature to develop brain tumor analysis methods using ML. The developed methods addressed the challenge as classification [1, 4–10], segmentation [11–16], and progression problems [2, 17–20]. Different algorithms are used in the developed methods, such as Convolutional neural networks (CNN) [9, 10, 12–14, 18], SVM [6], Long-Short Term Memory (LSTM) [18], XGBoost [11], k-means [5], ResNet [7], AlexNet [8], VGG [4, 19], and InceptionV3 [1]. Table 1 summarizes the reviewed related works.

Anaya et al. [1] developed a Radiomic ML classifier for differentiating Glioblastoma and solitary brain metastases. Their method extracts quantitative features from medical images to characterize tumors. Saba et al. [4] proposed a method that integrates hand-crafted features with a fine-tuned VGG-19 model. Tseng et al. [11] developed an optimized approach using XGBoost, feature selection, and image processing to tackle the complexities of tumor structure and MRI data noise. Islam et al. [5] integrated superpixels and Principal Component Analysis with a Template-based K-means algorithm to enhance brain tumor detection. Bhagat et al. [6] developed an SVM-based brain tumor classification method. Their method adopts the speeded-up robust feature algorithm for feature extraction and a swarm-based grasshopper optimization algorithm for feature selection. Kalaiselvi et al. [10] developed a CNN-based method for classifying benign tumor slices from MRI brain volumes. Integrating K-means clustering and discrete wavelet transform with SVM has demonstrated effectiveness in brain tumor detection and categorization [15]. Also, Kaur et al. [16] employed fuzzy C-means clustering for segmenting noisy MRI and CT scan brain images. Purwar et al. [20] proposed feature-based indexing algorithms for brain tumor MR images to facilitate efficient comparison and analysis.

Several researchers have emphasized the importance of preserving spatial information and addressing multi-scale lesions in brain tumor analysis [7, 12]. Furthermore, several deep learning methods are used in neuro-oncology for tracking tumor progression [8, 17, 18]. For glioma grading, the significance of precise tumor segmentation and overall survival prediction is further illustrated by Sun et al. [13]. In contrast, Pei et al. [9] introduced a context-aware deep learning method for brain tumor analysis, making a substantial contribution to the field.

Table 1 Summary of recent brain tumor analysis methods

Ref.	Year	Method	Goal	Dataset
[1]	2023	InceptionV3	Classification	BTD MRI-D
[4]	2020	VGG-19		BRATS
[5]	2021	K-Means		Their own
[6]	2022	SVM		Their own
[7]	2019	ResNet		Their own
[8]	2020	DCGAN AlexNet		..
[9]	2020	3D CNN		BRATS
[10]	2020	CNN		BRATS
[11]	2023	XGBoost	Segmentation	..
[12]	2020	DCNN		BRATS
[13]	2019	3D CNN		BRATS
[14]	2022	CNN		BRATS
[15]	2020	K-Means DWT		Their own
[16]	2017	Fuzzy C-mean		Their own
[17]	2019	..	Progression	Their own
[18]	2018	CNN-LSTM		Their own
[19]	2020	VGG16		Their own
[20]	2020	..		Their own

Table 2 Description of datasets per phase

Stage	Datasets	Description
Classification	Kaggle	7000 images from the Figshare, SARTAJ, and Br35H datasets
Segmentation	BraTS Challenge 2020	Glioma patients undergo multimodal MRI scans, including T1-weighted to T2-weighted FLAIR sequences
Progression	CIAr [21]	8,800 images from 20 newly diagnosed glioblastoma patients

3 NeuroSight

NeuroSight leverages advanced deep-learning methods for brain tumor detection, segmentation, and tracking.The proposed solution (Algorithm 1) comprises four main phases: (1) Data Preparation, (2) Brain Tumor Classification, (3) Brain Tumor Segmentation, and (4) Progression Tracking.

Algorithm 1 NeuroSight (*preMIR*,*curntMIR*)

Normalize and resize $preMIR$ and $curntMIR$
if EfficientNet($preMIR$) = 1 OR EfficientNet($curntMIR$) = 1 **then**
  $pt_segment \leftarrow$ AGSE-VNet($preMIR$)
  $ct_segment \leftarrow$ AGSE-VNet($curntMIR$)
  $progression \leftarrow$ Volume($ct_segment$) - Volume($pt_segment$)
  return $progression$
else
  return No Tumor is detected
end if

The training of NeuroSight involves a combination of several datasets to train the models at each phase (Table 2). The classification model is trained on a dataset from Kaggle[1] that includes four classes: glioma, meningioma, no tumor, and pituitary. Due to discrepancies in glioma categorization in the SARTAJ dataset affecting model training and external validations, these images were replaced with Figshare glioma images. The segmentation model utilizes data from the BraTS 2020 Challenge,[2] a widely respected dataset for brain tumor segmentation. Each scan is paired with a meticulously segmented ground truth label, which expert neuroradiologists uphold as a gold standard. The progression tracking system is evaluated using a dataset from The Cancer Imaging Archive (CIAr) [21]. Patients had two MRI scans during and post-treatment, featuring various sequences (Fig. 1). Approximately 25 images per scan, including Binary Tumor Masks, offer a comprehensive tumor view. Non-informative MRI sequences were omitted for NeuroSight's focus on pertinent data.

NeuroSight utilizes the EfficientNet model, a computationally efficient CNN for multiclass brain tumor classification, known for balancing accuracy and size [22]. EfficientNet's compound coefficient scales depth, width, and resolution, achieving higher accuracy with fewer parameters than alternatives. Its multi-scale feature extraction is adept at detecting subtle patterns in 3D MRI images, and resolution scaling enables NeuroSight to handle diverse input resolutions.

MRI scans that contain a tumor pass to the brain tumor segmentation phase. NeuroSight employs an improved version of the V-Net model called AGSE-VNet [23], designed for 3D medical image tasks, that directly handles volumetric data, capturing tumors' irregular shape and continuity across slices. V-Net utilizes 3D convolutions and deconvolutions in an encoder-decoder structure with skip connections, preserving spatial details. The Dice coefficient as a loss function addresses class imbalance. AGSE-VNet enhances V-Net with a squeeze and excite module for channel relationships and an attention guide filter module for attention, significantly refining segmentation by automatically enhancing relevant information and eliminating unnecessary details, such as noise.

The ultimate goal of NeuroSight is to monitor brain tumor progression over time, which is crucial for shaping treatment strategies and predicting outcomes. After brain tumor segmentation, the progression tracking phase involves two main stages:

1. Tumor volume calculations in NeuroSight involve counting voxels within the segmented region in 3D MRI

[1] https://www.med.upenn.edu/cbica/brats2020/data.html. Access Date: 1/11/2023.

[2] https://www.med.upenn.edu/cbica/brats2020/data.html. Access Date: 1/11/2023.

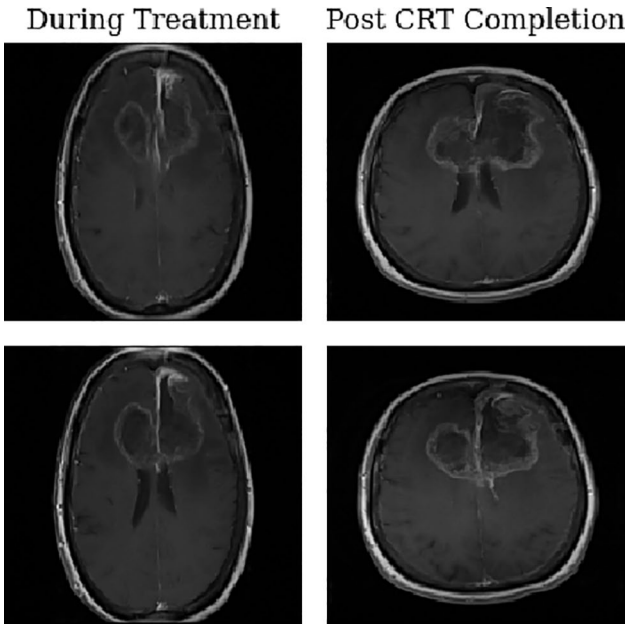


Fig. 1 Sample of the Progression Dataset

Table 3 Results of classification methods

Method	Model	Class	Spec.	Sens.	F1
Baseline	SVM	G	0.95	0.90	0.92
		M	0.86	0.91	0.88
		B	0.97	0.96	0.97
		P	0.96	0.95	0.96
	DT	G	0.72	0.66	0.69
		M	0.62	0.68	0.65
		B	0.90	0.89	0.90
		P	0.79	0.80	0.80
	XGB	G	0.97	0.85	0.91
		M	0.86	0.91	0.88
		B	0.98	0.99	0.98
		P	0.94	0.98	0.96
Recent	IncepV3	G	0.98	0.98	0.98
		M	0.97	0.96	0.97
		B	0.99	0.97	0.99
		P	0.99	0.98	0.98
	DNet121	G	0.98	0.98	0.98
		M	0.96	0.95	0.95
		B	0.99	1.00	1.00
		P	0.99	0.99	0.99
NeuroSight		G	0.98	0.99	0.98
		M	0.99	0.93	0.97
		B	0.99	1.00	0.99
		P	0.99	0.99	0.99

Table 4 Results of segmentation methods

Model	Region	DSC	Hausdorff95	Sensitivity	Specificity
3D-UNet	ET	0.60	46.96	0.66	0.99
	WT	0.82	10.70	0.86	0.99
	TC	0.74	11.63	0.75	0.99
Diff-UNet	ET	0.76	39.84	..	..
	WT	0.92	25.88	..	..
	TC	0.86	35.96	..	..
NeuroSight	ET	0.70	52.50	0.66	0.99
	WT	0.81	11.92	0.80	0.99
	TC	0.76	37.60	0.67	0.99

scans, quantitatively measuring tumor size for current and previous scans.

2. NeuroSight compares the current tumor volume with past scans to monitor size and shape changes, enabling precise tracking of tumor growth, shrinkage, or stability. This information is crucial for doctors to assess treatment effectiveness and plan future actions.

Furthermore, NeuroSight can help predict the patient's trajectory based on the observed trend. For example, a steadily increasing tumor volume may suggest more aggressive disease progression, prompting a change in the treatment strategy. This insight empowers medical professionals to adapt treatments based on observed tumor behavior, enhancing decision-making for better patient outcomes.

4 Experiments and results

NeuroSight involves three components: a tumor classifier, a tumor segmenter, and a progression tracker. Therefore, the evaluation process includes three experiments. Each component of NeuroSight is compared with other similar methods. The performance of EfficientNet is compared to SVM, Decision Trees (DT), XGBoost (XBG), InceptionV3 (IncpV3) [1], and DenseNet-121 (DNet-121) [24]. The AGSE-VNet is compared to 3D-UNet [25] and Diff U-Net [26].

The first experiment assesses the capability of the tumor classifier, EfficientNet, to classify MRI images into four classes: glioma (G), meningioma (M), benign (B), and pituitary (P). The EfficientNet model is evaluated using specificity, sensitivity, and the F1-Score. Table 3 presents the brain tumor classification component results. The EfficientNet model performed best in classifying the MRI images into pituitary. The relatively worst performance is achieved when classifying the MRI images into Meningioma. In general, the tumor classifier of NeuroSight, EfficientNet, outperforms the alternative methods. The

Table 5 Volume change and the rate of change results for neuroSight

ID	Previous	Current	Vol. Change	% Change
1	11350 Voxels	12361 Voxels	1011 Voxels	8.91%
2	10582 Voxels	10890 Voxels	308 Voxels	2.91%
3	10343 Voxels	9980 Voxels	−363 Voxels	−3.51%
4	9223 Voxels	8954 Voxels	−269 Voxels	−2.92%

EfficientNet improved the specificity, recall, and F1-Score up to 32.83%, 30.73%, and 31.3%, respectively.

The second experiment evaluates the tumor segmentation method, AGSE-VNet, across three delineated subregions: Enhancement Tumor (ET), Whole Tumor (WT), and Tumor Core (TC). The segmentation model is tested on high-grade and low-grade glioma cases with various MRI modalities (T2, FLAIR, T1, T1Gd). Five evaluation measures are used: Dice Coefficient (DSC), Hausdorff Distance (Hausdorff95), specificity, and sensitivity. DSC measures twice the area of overlap between the predicted segmentation and the ground truth. Hausdorff95 measures the extent to which each point of a model set lies near some point of an actual set and vice versa. Hausdorff distance is sensitive to outliers as it is determined by the point pair with the maximum distance. Thus, the 95th percentile of these distances, Hausdorff95, is used to ensure robustness against outliers. Table 4 shows the experimental results obtained using the segmentation methods. On average, AGSE-VNet achieved 2% and 155.8% better DSC and Hausdorff95 results than the alternative methods for the considered regions, respectively. It is worth mentioning that the authors of 3D-UNet and Diff-UNet published the presented results. Therefore, the recall and Specificity results of the Diff-UNet model are not provided by the authors [26].

The third experiment assesses the progression tracker components. The volume change and the rate of change measures are used in the experiments. Volume change is the difference in tumor volume (computed from the segmented MRI scans) between two time points. The rate of change is the fraction of the volume change by the time interval between the two scans. Table 5 shows the obtained results from applying the progression tracker to four patients. It is worth mentioning that medical professionals have meticulously validated these determinations.

To the best of our knowledge, the progression tracker component of NeuroSight is the first brain tumor progression tracking method based on tumor volume analysis over multiple time intervals. The reviewed tumor progression methods (Table 1) introduced various definitions of tumor progression, but neither tracked tumor progression over multiple time intervals. However, the concept of tumor progression of NeuroSight is achieved by using a quantified binary mask, offering an insight into the temporal evolution of tumors.

5 Conclusion and future works

This paper introduced NeuroSight, a novel brain tumor progression method consisting of three components: Tumor Classifier (EfficientNet), Tumor Segmenter (AGSE-VNet), and Progression Tracker. EfficientNet is employed for classifying 3D MRI images into four tumor classes, while AGSE-VNet handles segmentation. The Progression Tracker analyzes segmented tumor regions across periods, measuring volume changes. NeuroSight aids medical practitioners in informed decision-making for treatment plans. To the best of our knowledge, the progression tracker component of NeuroSight is the first brain tumor progression tracking method based on tumor volume analysis over multiple time intervals. Several experiments are conducted to test and compare the performance of NeuroSight with alternative methods. The utilization of EfficientNet leads to enhancements in specificity, recall, and F1-Score, with improvements of up to 32.83%, 30.73%, and 31.3%, respectively. The results showed 2% and 155.8% better Dice Coefficient and Hausdorff Distance using AGSE-VNet compared to alternative methods. Future extensions of NeuroSight could involve a prediction of tumor severity and patient survivability based on the tumor characteristics. Further improvements may include a detailed volume analysis of each part of the tumor for a more comprehensive understanding.

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Data availability The data that supports the findings of this study are: (1) classification data is openly available on Kaggle at https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset; (2) segmentation data is available on the multimodal brain tumor segmentation (BraTS) challenge 2020 at https://www.med.upenn.edu/cbica/brats2020/data.html; and (3) progression data was requested from The Cancer Imaging Archive [21].

Declarations

Conflict of interest No author associated with this paper has disclosed any potential or pertinent conflicts that may be perceived to have impending conflict with this work.

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